

Jennifer Bratburd

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EDUCATION

University of Wisconsin-Madison

Doctor of Philosophy (PhD) - Microbiology

Madison, WI

08/2014 - Present

University of California, Berkeley

Bachelor of Science (BSc) - Microbial Sciences, with honors

Berkeley, CA

08/2010-05/2014

SCIENCE OUTREACH AND ADVOCACY

Policy Engagement Coordinator

Madison, WI

Catalysts for Science Policy

08/2017-2018; 08/2019-Present

- Wrote and published policy memo on improving forensic science in Wisconsin with a team of graduate students
- Organized panel of experts to inform the public about issues in forensic sciences
- Engaged public with informal presentation on forensic science using examples from television
- Successfully obtained funding for a science policy podcast "In a Perfect Policy"
- Organized team of graduate students to evaluate and compare departmental graduate handbooks to find areas for improvement on mentorship policies.

Outreach Coordinator

Madison, WI

Catalysts for Science Policy

08/2016 - 08/2017

- Created interactive science policy outreach activities
- Organized volunteers at public science festivals
- Assisted in planning and publicizing events targeted graduate students interested in policy

Wisconsin Idea STEM Fellow

Madison, WI

University of Wisconsin, Madison

01/2016 - 05/2016

- Designed activities based on current research to present at outreach events

Outreach Volunteer Coordinator

Madison, WI

Currie Lab

08/2015 - 08/2018

- Coordinated undergraduate and graduate volunteers to engage visitors with hands-on demonstrations, tours, and games
- Conveyed research to various audiences at the Wisconsin Science Festival, Science Expeditions (families and general public), Expanding Your Horizons (middle school girls interested in science), and PLATO Frontiers in Life Sciences (retirees)

RESEARCH EXPERIENCE

Currie Lab, University of Wisconsin, Madison

08/2014 - Present

PhD Candidate

- Conducted research on microbial symbionts enabling pathogen resistance with human microbiomes in a germ-free mouse model and in defensive symbionts in fungus-growing ants
- Analyzed large-scale data genomic sequencing and metabolomic data

- Presented findings at professional society meetings including American Society for Microbiology and American Association for the Advancement of Science

Almeida Lab, University of California, Berkeley

02/2012 - 05/2014

Undergraduate Research Apprentice

- Assessed impact of drought stress on vector viral acquisition for honors research thesis
- Determined transmission efficiency of Grapevine Leafroll Virus-3 with different insect vectors
- Extracted RNA from grape petioles and insect to test for viral presence

University of California Education Abroad, Monteverde Institute

03/2013 - 05/2013

Student Field Researcher

- Cultured fungal and bacterial foliar endophytes from *Cecropia* trees to assess diversity

TEACHING EXPERIENCE

Microbiology 450: Diversity, Ecology & Evolution - Guest Lecturer

11/2017, 11/2018

- Lectured on new techniques in the field for large undergraduate class

Microbiology 305: Critical Analyses in Microbiology - Teaching Fellow

Spring 2017

- Developed teachable unit including lecture, discussion questions and homework
- Promoted learning through discussion between undergraduate students, focusing on critical thinking skills

Medical Microbiology and Immunology 341: Immunology - Teaching Assistant

Fall 2015

- Calculated grades and implemented opportunities for student questions to facilitate participation
- Developed lecture on mucosal immunity

WORK EXPERIENCE

ISA Summer Empowerment Internship

Tokyo, Japan

Student Facilitator

07/2014 - 08/2014

- Taught Japanese high school students about globalization and leadership
- Engaged students in discussion and facilitated presentations in English

Lab Assistant

Berkeley, CA

UC Berkeley, Dr. Ellen Simms Lab

05/2012 - 01/2013

- Assisted large-scale greenhouse experiment comparing host-symbiont interactions between multiple legume and *Rhizobia*
- Harvested *Lotus* nodules
- Scarified and germinated seeds under sterile conditions

Lab Assistant

Berkeley, CA

UC Berkeley, Dr. Michael Eisen Lab

03/2011 - 05/2012

- Collected fly behavior data to study fly-yeast interactions

PUBLICATIONS

J.R. Bratburd, R.A. Arango, H.A Horn. 2020. Defensive Symbioses in Social Insects Can Inform Human Health and Agriculture. *Frontiers in Microbiology*. <https://doi.org/10.3389/fmicb.2020.00076>

M.G. Chevrette*, **J.R. Bratburd***, C.R. Currie, R.M. Stubbendieck. 2019. Experimental Microbiomes: Models Not to Scale. *mSystems*. 10.1128/mSystems.00175-19

J.R. Bratburd*, C. Keller*, E. Vivas, E. Gemperline, L. Li, F.E. Rey, C.R. Currie. 2018. Gut microbial and metabolic responses to *Salmonella enterica* serovar Typhimurium and *Candida albicans*. *mBio* 9:e02032-18. <https://doi.org/10.1128/mBio.02032-18>.

J. R. Bratburd, A. Kowalkowski, G. K. Sidhu, K. Crocker. 2018. Establishing a Forensic Science Commission in Wisconsin. *Journal of Science Policy and Governance*. 13(1)

G. K. Blaisdell, S. Zhang, **J. R. Bratburd**, K. M. Daane, M. L. Cooper, R. P. P. Almeida. 2015 Interactions Within Susceptible Hosts Drive Establishment of Genetically Distinct Variants of an Insect-Borne Pathogen. *J Econ Entomol*. 108(4): 1531–1539doi: 10.1093/jee/tov153

* Indicates co-first authors

SEMINARS

“Of Gnotobiotic Mice and Men: Variation in Gut Microbiota Resistance to *Salmonella*.” Microbes in Health and Disease, UW Madison. December 4, 2018

“Exploring Defensive Symbiosis in the Human Gut Microbiome.” Microbiology Doctoral Training Program Seminar, UW Madison. April 11, 2018

“Human Gut Microbiome Community and Metabolic Response to Pathogens in a Gnotobiotic Mouse Model.” Microbes in Health and Disease, UW Madison. April 3, 2018

“The Metabolomic Response of the Gut Microbiota to Pathogen Infection.” Microbiology Doctoral Training Program Seminar, UW Madison. September 7, 2016

“Effect of Water Stress on Vector Transmission of Grapevine Leafroll-associated Virus-3.” College of Natural Resources Honors Symposium, Berkeley, California; May 2014

“Bacterial Endophytes of *Cecropia* Trees with an Ant Mutualism.” UC Education Abroad Research Symposium, Monteverde, Costa Rica; June 2013

HONORS AND AWARDS

Michael and Winona Foster Fellowship	07/2019 - Present
Microbes in Health and Disease Fellowship	08/2017 - 07/2019
Lasker Essay Competition Honorable Mention	06/2017
Perlman Poster Competition - 3rd Place	04/2017
Discovery Challenge Research Award Finalist	10/2016
College of Natural Resources Honors	05/2014
Sponsored Projects for Undergraduates Program	04/2013
Gilman Scholarship	11/2013

UNIVERSITY SERVICE

Admissions Committee, Microbiology Doctoral Training Program	11/2017
• Evaluated 239 total applications for graduate program	
Steering Committee, Microbiology Doctoral Training Program	08/2016

- Advocated for graduate student stipend raise
- Represented graduate student perspective for proposed policy changes